The Trade-off Was in the Labels: Causal Supervision for Turn-Aware Streaming ASR

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Abstract

Every voice agent must decide, in real time, whether the user has finished talking. The deployed default decides from acoustics alone — a voice-activity detector plus a silence timeout — and frontier commercial recognizers have instead moved the decision into the ASR model, where the words are. Every member of this *turn-aware* class is closed, and no training recipe for the capability has been published. We present, to our knowledge, the first open turn-aware streaming ASR system with a published training recipe: one small model — a LoRA adapter on Qwen3-ASR-0.6B, trained in hours on one GPU — that transcribes, detects end-of-turn semantically, handles dictation, and grounds transcription in session context. On a deployment-matched streaming benchmark it reaches 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute, replicated on a fresh test set drawn after development ended — an operating point no silence timeout reaches at any setting. The recipe rests on one principle: every label for a streaming decision must be computable from the input up to the decision point. Supervision inherited from offline corpora violates this systematically — clips are cut at exactly the boundaries a streaming model must detect, and transcripts record who spoke next — and we show that such *clairvoyant* labels do more than add noise: across eight controlled supervision compositions they manufactured training oscillation and an apparent recall-versus-precision frontier, which a one-variable intervention exposed as an artifact (appending one second of silence to the evaluation clips raised a "broken" model's fire recall from 0.10 to 1.00). A synthetic study isolates the mechanism with a tunable clairvoyant-label fraction. The failure class is not specific to time: a context prefix supplied to bias transcription became a copyable shortcut, and a wrong-user profile wrote the wrong entity into the transcript 40% of the time; counterfactual training examples in which context and audio deliberately disagree drive that intrusion to 0.8% while retaining most of the +28.9 pp entity-recall benefit. Weights, recipe, and benchmark are released.

Code: `https://github.com/19PINE-AI/turn-aware-asr`
Website: `https://01.me/research/turn-aware-asr`

1 Introduction

A voice agent that cannot tell when you have finished speaking is unusable in two symmetric ways: it interrupts you mid-thought, or it stares at you after you stop. The industry default delegates this decision to a cascade — a voice-activity detector (VAD) plus a fixed silence timeout, feeding a separate recognizer — and the cascade's problem is structural. Humans exchange turns with median gaps around 200 ms [Stivers et al., 2009, Levinson and Torreira, 2015], deployed timeouts sit at 500–1000 ms [Skantze, 2021], and pauses *inside* turns routinely exceed pauses between turns, so no threshold is both fast and safe (§2). The information that resolves the ambiguity — is this thought complete? — is in the words, and the words live inside the recognizer. Frontier commercial systems have drawn the conclusion and moved end-of-turn into the ASR model itself: Deepgram's Flux emits end-of-turn events from the recognizer's forward pass [Deepgram, 2026], and OpenAI's Realtime API waits longer after a trailing "ummm" than after a completed sentence [OpenAI, 2025]. We call this class **turn-aware streaming ASR**: one model that transcribes while deciding, from semantics and silence jointly, whether the turn is over. Every commercial member of the class is closed. The open landscape offers pieces — open weights with no training account (Kyutai STT, Parakeet-EOU), open turn *detectors* that do not transcribe (Smart Turn, LiveKit, TEN) — but no recipe for the combination (§8).

This paper presents, to our knowledge, the first open member of the class, with the complete training recipe. The system is deliberately small: a LoRA adapter [Hu et al., 2022] on Qwen3-ASR-0.6B [Shi et al., 2026], trained in hours on one GPU from ∼20k synthesized examples, that transcribes 0.5 s audio chunks incrementally, emits an in-transcript end-of-turn token when the turn is over, holds through dictated digit strings, and grounds transcription in a session context prefix. On a deployment-matched streaming benchmark it reaches 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute — 0.92 recall at matched latency and false fires on a fresh test set drawn after all development ended — an operating point outside the entire silence-timeout family on both sets (Figure 1; §5), and one that none of the open turn-aware systems reaches under the identical protocol. The context prefix lifts entity recall by $+28.9$ pp on entity-dense earnings-call audio while nudging WER down. Weights, recipe, and benchmark are released.

The recipe's substance is not architectural; the entire turn capability comes from the labels, and the paper's central finding is about what those labels must satisfy. Supervision inherited from offline corpora defines end-of-turn using audio *after* the decision point: offline clips are cut by forced alignment at exactly the boundaries a streaming model must detect, and offline transcripts record who spoke next. A streaming model can never see that audio. We call such labels *clairvoyant* (§3.1), and we show they do more than add noise: because the dependence is systematic, they partition the training data into internally consistent pools with incompatible optima, which manufactures both training-time oscillation between behavioral modes and an apparent recall-versus-precision trade-off across checkpoints. Our development record is the evidence (§6): eight successive supervision compositions — identical architecture, adapter, and data volume — traced what looked like a fundamental frontier, with every checkpoint either firing eagerly mid-utterance or barely firing at all. A one-variable intervention exposed the frontier as an artifact: appending 1.0 s of silence to the evaluation clips — restoring the future that a live microphone always delivers — took a "broken" checkpoint's fire recall from 0.10 to 1.00. One causal labeling rule — fire iff the words so far are semantically complete *and* at least 0.3 s of silence has been observed — plus a minimal-pair construction that forces the model to key on observables (§3.2) makes training monotone, with nothing else changed, and yields a single checkpoint that dominates all eight predecessors.

The principle is not specific to time. The same failure recurred on the context axis: trained only on examples where the biasing prefix matched the audio, the model learned to *copy* the prefix rather than listen, and a wrong-user profile wrote the wrong entity into the transcript

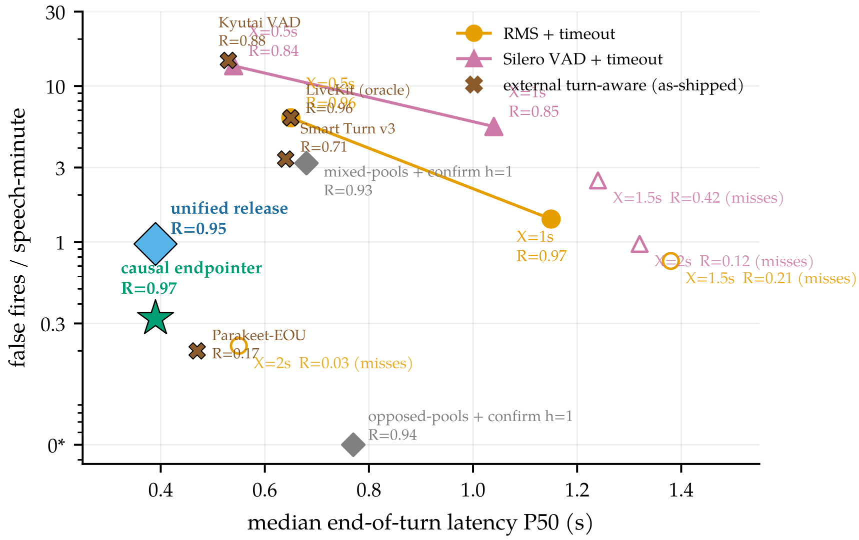

Figure 1. The causal model escapes the timeout curve. Median end-of-turn latency vs. false fires (log scale; 0* plotted at 0.05) on identical audio, detector, and scoring (§4.1). A silence timeout can only slide along its curve; the causally supervised model sits strictly inside the whole family because it reads completeness, firing 0.39 s after a finished thought while holding through a 1.4 s mid-sentence pause.

40% of the time. The leak has the same shape — the training target depended on a signal the model cannot trust at decision time — and the same repair: counterfactual examples in which context and audio deliberately disagree and the target follows the audio, which drive intrusion to 0.8% while largely preserving the biasing benefit (§3.3, §5). Two independent instances of one failure class, each cured by the same construction, ground a checklist for any streaming decision trained from offline logs (§7).

Our contributions:

1. **The first open turn-aware streaming ASR system, with its training recipe and benchmark** (§3–§5): one labeling rule and two pair constructions yield a single small model that transcribes, emits in-transcript end-of-turn events, holds through dictation, and grounds transcription in context, beating every silence-timeout setting; weights, recipe, and a deployment-matched benchmark — under which the historical model ranking *inverts* — are released.

2. **A failure class, with a diagnosis by intervention** (§3.1, §6): clairvoyant labels — streaming supervision that depends on post-decision input — manufacture training oscillation and phantom capability trade-offs; a one-variable experiment falsifies the frontier that eight supervision compositions had appeared to trace, and a synthetic study isolates the mechanism with a tunable clairvoyant-label fraction.

3. **Evidence that the principle generalizes** (§3.3, §5): the contextual analog of the temporal leak (context copying; 40% wrong-profile intrusion) is cured by the analogous counterfactual construction (intrusion to 0.8%, biasing intact), folding dictation and context grounding into the same checkpoint; both instances distill into a checklist for streaming supervision pipelines (§7).

2 The task, and why a threshold cannot solve it

Two turns bound the problem from opposite sides. The first is a pause that means *wait*: "My number is four one five ... (0.9 s) ... five five five ... (1.1 s) ... zero one nine two." A 0.7 s timeout fires twice mid-number; the silences carry no evidence that the turn is over, but the words do — six digits of a ten-digit pattern predict continuation — and production teams report exactly this failure [LiveKit, 2026]. The second is a completion that means *go*: "Hello? I'm still here." The thought is complete the instant the last word ends, and a human would respond within ∼200 ms [Stivers et al., 2009]; a 1.0 s timeout makes the user wait five times that long, for nothing. The first case needs the system to hold *through* long silence, the second to fire at near-zero silence, and no single threshold can do both, because the gap between the cases is not measurable in seconds of quiet. On our own conversational benchmark the silence-gap distributions of within-turn pauses and between-turn boundaries overlap almost completely, and the pauses are the *longer* of the two (§6); Skantze [2021] documents the same overlap across the dialogue literature, and the resulting no-good-threshold dilemma has defined cascade endpointing since Raux and Eskenazi [2008]. The resolving signal is semantic completeness, which is exactly what a recognizer computes on the way to a transcript. §4.2 builds probes for these two scenarios; §5 closes them with training data alone.

Transcription has its own version of the argument. A voice agent hears names, tickers, account emails — tokens nearly unrecoverable from acoustics alone but obvious given who is calling about what. Every major commercial STT ships a context mechanism, yet none publishes controlled uplift numbers, and the open model we build on describes its context facility in one qualitative sentence [Shi et al., 2026]. A turn-aware model with a context slot can ground the whole session's transcription in the profile the agent already has; we measure exactly this in §5.

The system under study. A single-channel audio stream arrives in 0.5 s chunks. At each chunk boundary the system must extend a running transcript and decide whether the turn has ended — an irreversible, latency-sensitive event. A text context prefix may be supplied at session start. We start from Qwen3-ASR-0.6B [Shi et al., 2026], an open unified speech model (AuT audio encoder + dense Qwen3 decoder) with strong offline WER but no endpointing, claim two reserved vocabulary slots as marker tokens <EAGER_END_SPEECH> and <END_SPEECH>, and LoRA-fine-tune [Hu et al., 2022] the decoder to emit them inside the transcript at endpoint moments (rank 16 for the endpointing experiments, 32 for the released unified model; adapter details in Appendix H). Training data is synthesized from LibriSpeech [Panayotov et al., 2015] and AMI [Carletta et al., 2005]; training takes hours on one GPU. At inference, three interpretable knobs sit outside the model, one in front and two behind (Figure 2): an RMS **energy gate** that never *starts* a segment on a silent chunk, a **confirm horizon** h that holds a fire candidate for h further silent chunks before signaling END, and a **max-segment force-flush** that bounds the committed transcript segment. Every reported arm states its knob settings.

3 Method: causal supervision

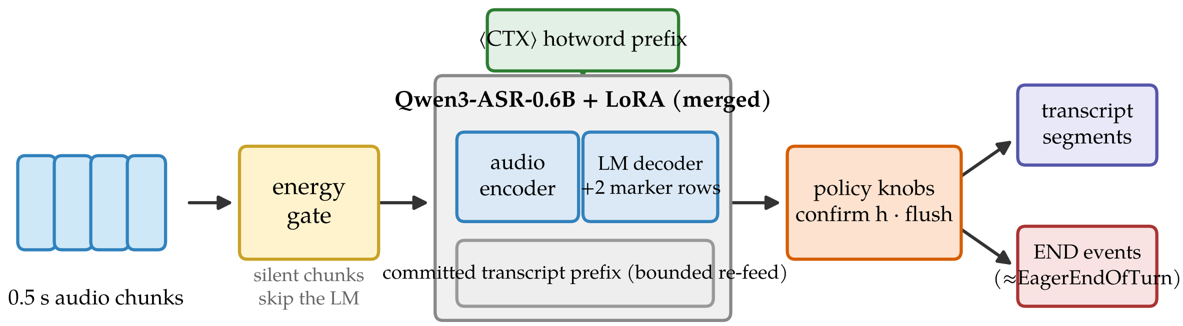

Figure 2. System shape. Audio chunks pass an energy gate; the merged LoRA model transcribes the current bounded segment with the `<CTX>` prefix in the system slot and may emit marker tokens; two policy knobs sit behind the model. Outputs are transcript segments plus END events, the same event shape as Flux's `EagerEndOfTurn`.

Table 1. The landscape (verified July 2026; discussion in §8). "Turn detectors" are separate models beside a recognizer; "cascade" is VAD + silence timeout + separate ASR, the deployed default and our measured baseline.

	This work	Flux / `sem.vad`	Kyutai STT	Parakeet-EOU	Turn det.	Cascade
Streaming ASR	✓	✓	✓	✓	—	✓
In-model end-of-turn	semantic	semantic	sem. head	token	sep. model	timeout
Open weights	✓	—	✓	✓	mostly	✓
Turn-training recipe	✓	—	—	—	Smart Turn	n/a
Context biasing, measured	✓ (+28.9 pp)	—	—	—	—	ASR-dep.

3.1 The constraint, and the failure class it forbids

A streaming model may act at time t only on the input prefix up to t. The constraint sounds too obvious to state, yet supervision built from offline corpora violates it systematically, because offline data encodes the future in two places. First, offline ASR corpora are segmented by forced alignment, so clips are cut at the end of speech; endpoint labels or evaluations built on such clips demand a fire on audio that ends *during or immediately after* speech — a condition deployment never presents, because on a live channel silence keeps arriving after a real speaker stops. Second, offline transcripts record who spoke next, so pause labels derived from segmentation (same speaker continues: disfluency, hold; another speaker follows: boundary, fire) are functions of the future; measured at the decision point, the two classes are statistically indistinguishable (§6). We give the failure class a name:

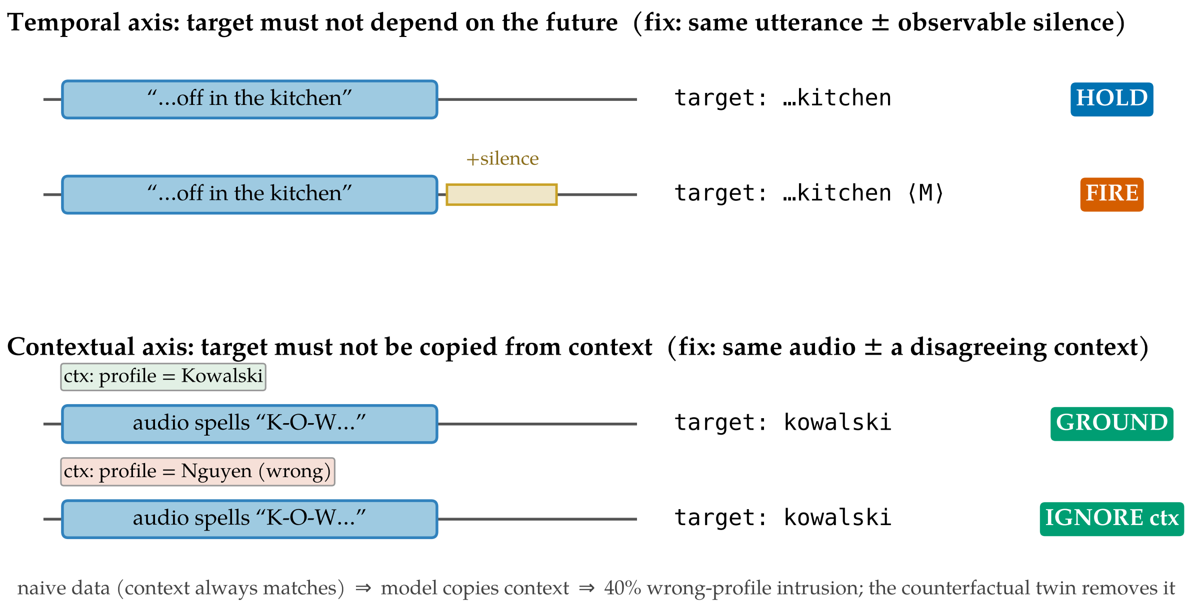

Figure 3. One failure class, two instances. Temporal axis (§3.2, §6): labels that peek at the future manufacture a phantom frontier; the fix is the minimal pair, the same utterance ± an observable silence tail. Contextual axis (§3.3): a context prefix that always matches the audio becomes a copyable shortcut; the fix is the counterfactual twin, the same audio with a matching and a disagreeing context, target following the audio.

Definition: clairvoyant labels

A label for a streaming decision at time t is *clairvoyant* if its value depends on input after t. Pools containing clairvoyant labels do not merely add noise: because the dependence is systematic, they partition the data into internally consistent subsets with incompatible optima. The observable symptoms are (a) training-time oscillation between mode attractors and (b) apparent capability trade-offs across checkpoints — a phantom Pareto frontier. **Principle: every training label and every evaluation target for a streaming decision must be computable from the input up to the decision point.**

Clairvoyance is target leakage transposed into time, and the class is wider than time: the same structure appears whenever the target correlates with any signal the model cannot *trust* at decision time. A context prefix that always matches the audio during training is such a signal — at deployment the profile can simply be wrong (§3.3). Figure 3 shows the two instances side by side. In both coordinates the cure is a construction rather than a loss term: minimal pairs or counterfactual twins that make the untrustworthy signal uninformative by design, so the only policy consistent with the training data is the one that keys on observables. §6 presents the evidence for the symptoms — the controlled eight-composition record, the intervention that falsified the frontier, and a synthetic study that isolates the mechanism.

3.2 The endpointing recipe

The entire turn capability comes from the labels. One rule generates every training example: *emit <EAGER_END_SPEECH><END_SPEECH> at a point iff the speech so far is semantically complete*

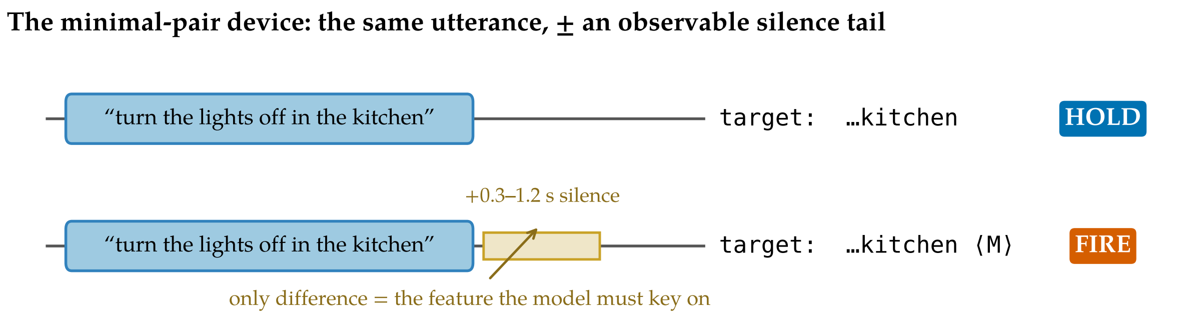

Figure 4. The minimal-pair device. Each complete utterance appears twice: without a silence tail (target: transcript only) and with a 0.3–1.2 s tail (target: transcript + marker). The only difference between the opposite-labeled examples is the observable the model must key on.

Table 2. The eight training schemas (∼12k examples; LibriSpeech and AMI roughly 50/50). M = <EAGER_END_SPEECH><END_SPEECH>.

#	Construction	Target
1	complete utterance + 0.3–1.2 s silence tail	`text` M
2	*same utterances as #1*, tail ≤0.1 s	`text`
3	speech truncated mid-utterance at 30–80%	partial `text`
4	complete A + gap 0.3–2.5 s + B (+ tail)	`A` M `B` M / `A` M `B`
5	*incomplete* A + gap + B + tail	`A B` M
6	complete utterance + long silence (2–4 s)	`text` M
7	pure silence 0.5–4 s	(empty)
8	leading silence 0.5–2 s + utterance + tail	`text` M

AND ≥0.3 s of silence has elapsed since the last speech. Both conditions are functions of the audio prefix. Completeness is decided by a transcript heuristic (trailing filler or connective, mid-word cut, or very short non-acknowledgment ⇒ incomplete; Appendix A); silence is decided by the synthesized waveform itself.

Two constructions carry the semantics-versus-silence discrimination that §2 asked for (Table 2). The **minimal pair** (schemas 1–2, Figure 4) presents the same complete utterance twice, with and without an observable silence tail, with opposite targets: completeness alone must not fire; the model has to *see* the silence. This is the direct antidote to supervision under which silence becomes optional (row 4 of the composition record, Table 7). The **pause pair** (schemas 4–5) presents two-utterance sequences whose gap is bridged or fired depending only on whether the words before the pause are complete: "…and then, *(pause)*" holds; "…that's everything. *(pause)*" fires, even if the speaker later resumes. Quick resumes after a legitimate fire become two transcript segments — a measured cost, not an error (§4.1) — because at decision time the correct answer does not exist on the channel.

Schemas 7–8 cover the silence-heavy channel shape that pure-utterance corpora never show, and pair examples mix same- and different-speaker sources, so the fire decision cannot key on speaker identity, which single-channel deployment never observes (Appendix A). An iso-count ablation (§5, Table 6) isolates what each construction buys.

3.3 The dictation and context constructions

The same discipline extends the recipe to the two transcription behaviors that §2 motivated. **Dictation:** partial phone numbers followed by a pause are labeled *hold*, because the pattern predicts continuation — the completeness half of the rule, instantiated for enumerations; complete numbers plus observed silence fire. **Spelled entities and context:** synthesized utterances spelling uncommon names and email addresses are targeted in normalized written form, half with a user profile in the context slot. These schemas extend the endpointing pool to $\sim$15k examples, and the same LoRA is retrained on the extended pool.

Matched-context supervision alone, however, reintroduces the §3.1 failure class on the context axis. If the profile always agrees with the audio, the target is predictable from the prefix without listening, and the model learns that the context *is* the answer — it copies whatever entity the prefix names (§5 measures the consequence: up to 40% wrong-profile intrusion). The **counterfactual twin** removes the shortcut: a third of the spelled examples carry a *conflicting* profile — the context names one identity, the audio spells another, and the target follows the audio — exactly as the minimal pair presents the same utterance with and without a silence tail. A second counterfactual of the same shape covers natural-speech biasing: the context is an entity list that disagrees with the audio, and the target again follows the audio. The released pool ($\sim$20k examples) folds in both counterfactuals plus a 20% plain-transcription replay fraction that anchors offline WER (Appendix F); the released adapter is rank 32, which §5 shows is what the added breadth costs.

4 Evaluation protocol

4.1 A deployment-matched streaming benchmark

Endpointing is evaluated by streaming replay, and the causality principle applies to the benchmark itself: an evaluation whose clips end at forced-alignment cuts demands behavior on a condition deployment never produces, and §6 shows such an evaluation actively misdirects development.

Material and ground truth. Continuous single-channel stretches from held-out AMI meetings: one target speaker's utterances at their true timeline offsets, silence between them, 30–60 s per stretch — the voice-assistant shape, where audio never ends at a speech boundary. Each utterance-end boundary is classified by what is observable *on this channel*: **turn-final** (same-speaker gap $\geq$2 s), where the system should fire; **continuation** (gap in $[0.3, 2)$ s, same speaker resumes), where firing is a measured segmentation cost rather than an error, because the correct answer depends on the future; and **internal** (pauses inside utterances, or $<$0.3 s after speech), where firing is a false fire. Metrics: boundary recall within $[-0.25, +1.5]$ s, false fires per speech-minute, latency percentiles, resume rate, streaming WER of flushed transcripts, and per-chunk compute. The development set has 25 stretches (96 turn-final boundaries); a 2$\times$-larger confirmation set (50 stretches, 184 boundaries) was drawn after all development ended, and a 100-stretch held-out set (384 boundaries) is used for the released model. An early version of the benchmark itself contained a clairvoyant rule — promoting boundaries to turn-final when another meeting speaker interleaved, which is inaudible on this channel; our own checklist (§7) caught it, and all numbers use the corrected classification (Appendix B).

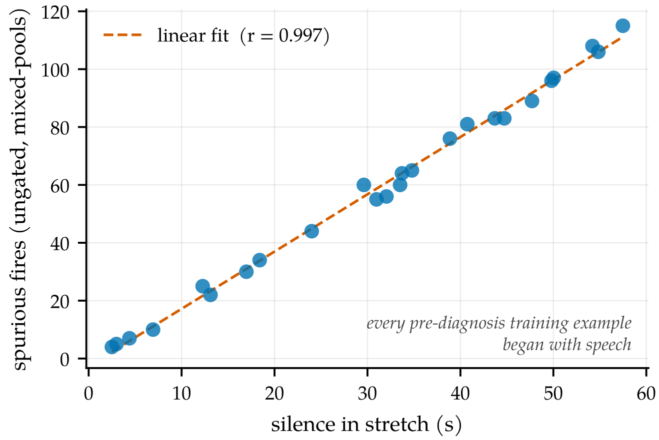

Figure 5. Silence incompetence. Spurious fires of the ungated mixed-pools model track the silence content of each stretch almost perfectly; ungated recall is therefore uninterpretable, and all reported arms run behind the energy gate.

Why comparisons run behind the energy gate. Models trained only on speech-initial examples hallucinate on silence: the ungated mixed-pools model sprays markers at ~1.9 fires per second of silence (r=0.997 against stretch silence content; Figure 5), which makes *ungated* recall uninterpretable — a random 1.9 Hz spammer scores 0.96 against the tolerance window. All comparisons therefore run behind the energy gate, and the causal recipe additionally bakes silence schemas into training.

4.2 Probes and offline evaluations

The two §2 scenarios get probes, both run through the exact streaming stack of §4.1. The **dictation probe** replays ten-digit phone numbers (3-3-4 groups from held-out Free Spoken Digit Dataset speakers, 0.6–1.2 s inter-group pauses) and scores premature fires per number, final-boundary recall, and digit accuracy. The **spelled-entity probe** decodes synthesized utterances spelling uncommon names and email addresses, with and without a user-profile prefix, plus a wrong-user *distractor* profile, and scores exact match and wrong-profile intrusion. Both probes were enlarged after development ended (250 sequences; 240 spelled items over 120 identities), and released-model numbers are quoted on the enlarged versions. **Context biasing** on natural speech is measured on Earnings-22 [Del Rio et al., 2022] — spontaneous earnings-call speech dense in tickers, executives, and products, with entities LLM-extracted and filtered to genuine proper nouns — as entity recall with and without a relevant hotword prefix. **Offline WER** is scored on the full official LibriSpeech splits, and **serving** is measured on stock vLLM rather than a simulator (Appendix G).

5 Results

5.1 Endpointing: one checkpoint dominates

Table 3 reports the main result. Against the strongest prior supervision compositions (the mixed-pools and opposed-pools models of Table 7, each at its best knob setting), the causal model with the gate alone is better on every axis at once: its *raw* false-fire rate beats the mixed-pools model's *confirmed* rate, at half the latency and higher recall. Against the deployed

Table 3. Main endpointing result (deployment-matched replay, energy gate on; the latency-vs-false-fire view is Figure 1). Rows 1–4: the 25-stretch development benchmark (96 turn-final boundaries). Last row: the fresh 2×-larger confirmation set (50 stretches, 184 turn-final boundaries) drawn after all development ended. WER is a fraction; means can exceed 1 through insertions. †WER mean dominated by long-monologue repetition loops when no force-flush bounds the segment (WER *median* 0.30 on both stretch sets; the force-flush of Appendix G bounds this tail).

Policy	Recall↑	P50↓	P95↓	False/min↓	WER_{mean}↓
mixed-pools + gate + confirm $h=1$	0.927	0.68 s	1.27 s	3.2	0.36
opposed-pools + gate + confirm $h=1$	0.938	0.77 s	1.06 s	0.0	4.96†
causal + gate (ours)	**0.969**	**0.39 s**	**0.70 s**	**0.3**	1.43†
causal + gate + confirm $h=1$	0.969	0.89 s	1.20 s	0.0	1.43†
causal + gate — fresh 50-stretch set	0.924	0.42 s	0.70 s	0.2	4.63†

cascade, the comparison is structural. A timeout must be long enough to survive within-turn pauses, so it can only trade latency against false fires along one curve; the causal model escapes the curve by reading completeness, combining exactly the pair of behaviors §2 argued no threshold can combine. The full timeout family, swept over both detectors and all settings on both stretch sets (Appendix C): matching the causal model's recall costs a timeout 3× the latency and ∼5× the false fires, and no setting reaches its latency-and-false-fire corner at any recall.

Two further observations. The inference crutches retire: the confirm horizon that earlier compositions needed to be usable cancels only 3 fire candidates for the causal model, versus 241 for the mixed-pools model — the phrase-versus-turn discrimination moved from inference policy into the weights. And the result survives fresh data: because the 25-stretch benchmark was reused during development, we drew a 2×-larger set after all development ended (Table 3, last row). Latency and false fires hold; recall dips 4.5 pp, a difference whose 95% interval includes zero (Appendix C), and we quote the fresh set whenever a single operating estimate is needed. One caveat by design: quick resumes after a legitimate fire are always segmented (resume rate 1.0), because the correct choice is unknowable at decision time; deployments preferring merged segments pay +0.5 s via the confirm horizon.

External turn-aware systems on the same protocol. We ran the open turn-aware systems of §8 through the identical benchmark — same audio, energy gate, and scoring — as-shipped, sweeping their public thresholds rather than retraining (Figure 1). None reaches the causal model's corner. Smart Turn v3 (an acoustic classifier triggered on VAD silence) sits at 0.71 recall and 3.4 false fires per speech-minute; LiveKit's text detector, handed an *oracle* transcript, reaches 0.96 recall but at 6.25 false fires; NVIDIA's Parakeet-EOU fires so rarely it holds recall to 0.17; Kyutai's semantic-VAD head reaches 0.88 recall only at 14.6 false fires. The detectors are gated on VAD silence by construction and slide along the timeout curve; LiveKit's oracle-transcript number shows the binding constraint is that gating, not transcript quality. To our knowledge this is the first measurement of the open turn-aware class on a common protocol; it is as-shipped, not best-achievable (none was tuned for AMI close-talk).

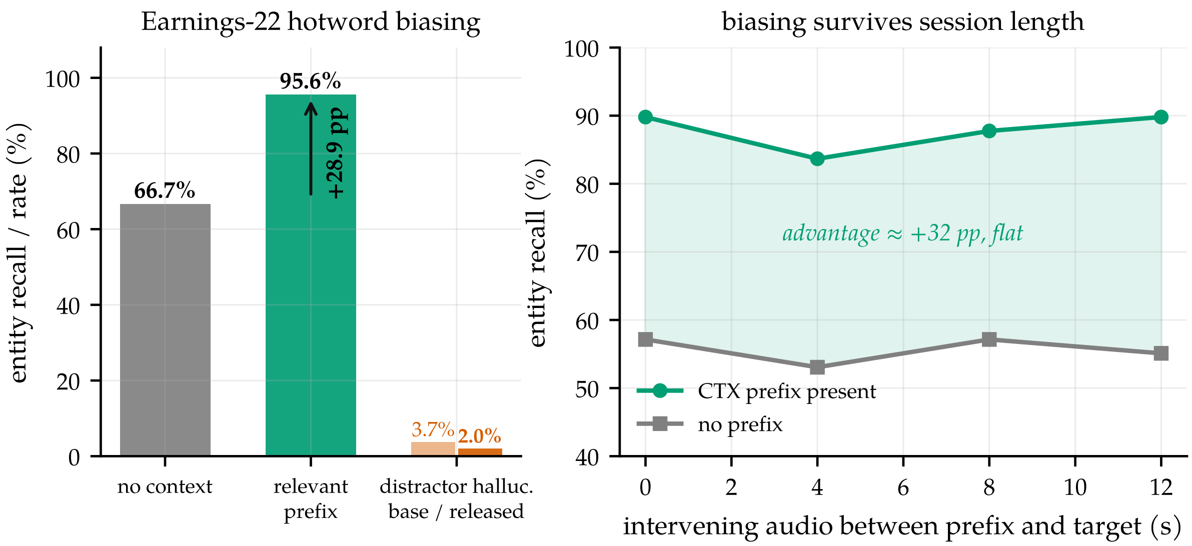

Figure 6. Context biasing, measured. (a) Earnings-22 base-model entity recall without context and with the relevant hotword prefix (+28.9 pp), and the distractor-prefix hallucination rate for the base model versus the released rank-32 model whose natural-speech counterfactual drives it from 3.7% to 2.0%. (b) Recall vs. seconds of unrelated speech between prefix and target: the advantage does not decay with session age.

5.2 Context biasing, dictation, and the copying result

Context biasing works, and survives session length. On Earnings-22, a relevant hotword prefix lifts the base model's entity recall from 66.7% to 95.6% (+28.9 pp) while nudging WER down (15.3→15.0%): the prefix acts as a domain hint, not a distortion (Figure 6a). With up to 12 s of unrelated speech interposed between prefix and target words, the advantage stays flat at ≈+30 pp (Figure 6b) — the property a voice agent actually needs, since the user's account entities must help at minute five, not just in the first utterance. The base model's technical report describes this capability in one qualitative sentence and publishes no numbers [Shi et al., 2026]; to our knowledge these are the first on this model family, and a concurrent benchmark corroborates the effect across commercial systems [Durmus et al., 2026].

Dictation, and the copying result. The conversational-only endpointing model of §5.1 fails both §4.2 probes exactly as §2 predicts: it reads a digit group as a finished thought (5.0 premature fires per dictated number, worse than a 0.5 s timeout) and spells emails at 4% exact (Table 4). The dictation schemas of §3.3 fix the former immediately. The context schemas, trained with matched profiles only, produce the predicted copying: the model writes whatever entity the prefix names, so a wrong-user profile puts the wrong email into the transcript — 11.3% wrong-profile intrusion, rising to 40% with longer training. Adding the counterfactual twin drives intrusion to zero at every checkpoint of the run on the training probe — 0.8% (95% CI $[0.2, 3.0]$%) on the enlarged post-development probe — while leaving dictation and matching-context accuracy intact (Table 4): context becomes a prior, not a substitute for listening.

Table 4. Dictation and spelled-entity probes. Premature fires per dictated 10-digit number, final-boundary recall, digit accuracy; exact-match on spelled names/emails without/with the profile; wrong-profile intrusion. Conversational-only and released rows are scored on enlarged probes (250 sequences; 240 spelled items over 120 identities); the intermediate row uses the smaller original probes. Confidence intervals and a scoring-window sensitivity analysis are in Appendix C. The conversational-only model's nonzero intrusion is the base model's native context-following, which that fine-tune leaves untouched.

	Premat./seq ↓	Final rec. ↑	Digit acc ↑	Name −/+	Email −/+	Intrus. ↓
conversational only	4.96	0.82	0.889	0.25 / 0.82	0.04 / 0.58	1.7%
timeout X=0.5 s	2.00	1.00	—	—	—	—
timeout X=1.0 s	0.74	1.00	—	—	—	—
+ dictation (match-ctx)	0.36	0.78	0.996	1.00 / 1.00	0.03 / 0.93	11.3%
released (r32, unified)	**0.16**	**0.88**	**0.998**	**0.98 / 1.00**	0.11 / **0.93**	**0.8%**

5.3 The released unified model

The release is the merged unified checkpoint, the label-spec builder and training code, the dictation and spelled-entity probes, and the replay benchmark.

One released model, four behaviors. The released checkpoint — a rank-32 adapter trained on one pool of ∼20k examples — endpoints conversation, holds through dictation (0.16 premature fires per number at 0.998 digit accuracy, better than every timeout setting), grounds spelled entities in context (names 100%, emails 93% exact), and ignores a wrong profile (0.8% intrusion), with no deployment-time checkpoint swap. The shipped dictation scoring window is in fact conservative: re-scored by each fire's offset from the true end of speech, the residual "premature" fires land within one chunk *after* the last digit, giving 0.07 genuine mid-sequence fires per number and zero late fires (Appendix C). On a large held-out replay set (100 stretches, 384 turn-final boundaries) the released model scores 0.982 boundary recall at 1.30 false fires per speech-minute, P50 0.38 s — still strictly inside the timeout family of Figure 1; Table 5 collects its endpointing numbers across stretch sets. Like the pure endpointer, it exposes the confirm-horizon dial: on the development set a one-chunk horizon takes it to zero false fires at 0.958 recall (P50 0.89 s) for deployments that prefer conservatism over latency. And the capability is a property of the recipe rather than a lucky checkpoint: retraining under three seeds gives premature-fire rate 0.16 ± 0.00, email-with-context accuracy 0.90 ± 0.07, and intrusion 0.000 ± 0.000 (Appendix C).

Table 5. The released unified checkpoint (rank-32) on the replay benchmark, both stretch sets, alongside the pure-endpointing causal checkpoint of §5.1. dev = 25-stretch development set (96 turn-final boundaries); held-out 100 = the 100-stretch set (384 boundaries); fresh 50 = the post-development confirmation set (184 boundaries).

Model	Set	Recall	P50	P95	False/min
released r32 + gate	dev	0.948	0.39 s	0.65 s	0.97
released r32 + gate + confirm h=1	dev	0.958	0.89 s	1.15 s	0.0
released r32 + gate	held-out 100	0.982	0.38 s	0.64 s	1.30
pure endpoint (causal, r16) + gate	dev	0.969	0.39 s	0.70 s	0.3
pure endpoint (causal, r16) + gate	fresh 50	0.924	0.42 s	0.70 s	0.2

Which frontiers are real. Folding four behaviors into one adapter costs endpointing precision: on the development set, 0.97 false fires per speech-minute versus 0.3 for the pure-endpointing checkpoint (Table 5), a genuine capacity trade-off we report and do not tune away (a deployment that only needs turn-taking can run the pure checkpoint). This is the one frontier that survives the diagnosis: the endpoint recall-versus-precision frontier dissolved when the labels became causal (§6), but the breadth-versus-precision cost at fixed adapter capacity does not — and it recedes with capacity, which is why the release uses a rank-32 adapter. At rank-32 the added replay and biasing schemas leave endpointing recall intact (0.95 versus 0.97 on the development set) rather than eroding it, as they did at rank-16. The context axis carries the same lesson. The spelled-entity counterfactual drives wrong-profile intrusion to near zero but does not, by itself, cover *natural-speech* distractors: an entity list naming companies absent from the audio is hallucinated 6.3% of the time. The natural-speech counterfactual of §3.3, folded into the released pool, cuts that hallucination to 2.0% (below the base model's 3.7%) while preserving a +28.2 pp entity-recall uplift, close to the base model's raw biasing capability. The residual cost is on absolute entity recall, which the anti-hallucination pressure lowers slightly; the aggressiveness of the counterfactual sets where on that recall-versus-hallucination curve the model sits.

5.4 Schema ablation

Retraining the endpointing recipe with one construction removed at a time isolates what each buys (Table 6; iso-count: the dropped quota is redistributed across the remaining schemas, holding volume fixed; each arm is selected on the holdout composite and scored on the 25-stretch replay set). Each construction has a distinct predicted failure, and two of three reproduce it. Removing the **minimal pair** (schema 2) makes silence optional again: gated false fires rise 1.08→5.28 per speech-minute and the confirm horizon cancels 34 candidates versus 10 — the pathology and its inference crutch both return. Removing the **silence schemas** (7–8) costs silence-competence: ungated, the model emits 1458 markers on silence versus 8, masked in the gated numbers by the energy gate. Removing the **pause pair**'s incomplete half (schema 5) does not raise false fires; the model is mildly more conservative (recall 0.948→0.906, false fires 1.08→0.43), so pause discrimination is largely covered by schema 4's complete-A gaps.

Table 6. Schema ablation on the endpointing pool (iso-count; 25-stretch replay, best checkpoint each). Each arm is an independent retrain selected on the holdout composite, so the baseline here (0.948/1.08) is a different checkpoint from the causal model of Table 3 (0.969/0.3). "Silence-fires" is the ungated marker count on silence; "h=1 cancels" is the confirm-horizon crutch metric.

Arm	Recall ↑	False/min ↓	Silence-fires ↓	h=1 cancels	Predicted failure?
baseline (full pool)	0.948	1.08	8	10	—
−minimal pair (#2)	0.906	5.28	5	34	yes (silence optional)
−silence (#7–8)	0.969	1.94	1458	12	yes (silence spray)
−pause pair (#5)	0.906	0.43	8	4	no (redundant)

5.5 Costs: offline WER, and serving

Offline cost, measured and attributed. The endpoint fine-tune costs +1.1/+2.7 pp WER on LibriSpeech test-clean/test-other (Appendix C, Table 12). The mechanical explanation —

marker fires truncating the decode — is falsified by intervention: banning the marker tokens at decode time drops fires to zero and does not move WER. The regression is distributional drift from a narrow-schema fine-tune; mixing plain transcription examples into the pool (the standard mitigation, exercised in the released model, which scores 3.5%/6.9%) recovers most of it, and Appendix F documents the full attribution, an AdamW weight-decay hazard for recipes that train a few vocabulary rows against a frozen embedding, and one negative result (Muon underperforms AdamW on these LoRA factors). Streaming WER, the deployment-relevant number, is unaffected.

Serving on stock vLLM. The model serves on stock vLLM [Kwon et al., 2023] at deployment-compatible cost. Three findings, detailed in Appendix G: bounded re-feed of the last audio window is the correct serving primitive (the natural per-chunk-placeholder shape is out of distribution for a model trained on single segments and destroys it, 88% versus 3.8% WER); median per-chunk compute is 21 ms on an otherwise-idle GPU and 84 ms under a co-tenant — 5.4× faster than the transformers simulator used during development, so latency claims made on $O(T^2)$ re-decode simulators overstate the cost of this approach by that factor; and per-frame P95 stays flat through 16 concurrent sessions on a fractional-GPU slice (77 ms at $N{=}16$, well inside the 500 ms budget), with a 20 s force-flush bounding both the WER and compute tails. The in-engine efficiency variant — windowed KV with the `<CTX>` prefix pinned as an attention sink [Xiao et al., 2024] — is validated in a companion systems paper [Li, 2026].

6 Analysis: how clairvoyant labels manufacture a trade-off

The claim that the labels are the load-bearing element of §3 rests on a controlled record. Before the causal rule, eight successive supervision compositions shared everything — architecture, adapter, data volume, source corpora — except the labels, and their record reads as a dose–response study of what contradictory supervision does to a model. This section presents that record, the intervention that diagnosed it, the mechanism behind the oscillation, a synthetic study that isolates the mechanism outside speech, and what the episode implies for evaluation.

6.1 The composition record

Table 7 compresses the record. The first composition supervised on offline clips alone: fire at every clip end. It scored perfectly offline and was unusable live, firing 89 times per speech-minute mid-utterance. Successive compositions added pools intended to fix this — truncated-speech holds, trailing-silence fires, hold pools on disjoint audio — and every addition moved the model *along* an apparent recall-versus-precision frontier, never off it. Two behaviors recurred across the family. Checkpoints were bimodal: eager ones fired mid-utterance, conservative ones missed turns, and no checkpoint did both jobs. And late compositions did not converge at all: holdout accuracy oscillated for the entire run between a *fire-mode* attractor (fire class ≈1.0, hold class ≈0.4) and a *no-fire-mode* attractor (fire ≈0.1, hold ≈1.0), as in Figure 7 (left). By the eighth composition (row 6) the working conclusion, recorded in the project log at the time, was that the two behaviors lay on a fundamental Pareto frontier, and the recommendation was to ship two checkpoints, one eager and one conservative. The remainder of this section dismantles that conclusion.

Table 7. The supervision-composition record. Eight compositions, one architecture, compressed into six rows: rows 1 and 2 each span two iterations (a data scale-up; a truncation-dose increase), and each later row adds one labeled pool to the mix above it, except row 6, which *replaces* row 5's same-audio hold pool with disjoint audio. "Early" latencies are negative: the model fires seconds before the turn ends. Row 7 is the causal recipe of §3.2.

#	Supervision change (cumulative)	Observed behavior
1	offline endpoint pool only: fire at every clip end	fires eagerly and "accurately" offline (100% fire recall); in streaming replay, fires 89× per speech-minute mid-utterance
2	+ truncated-speech hold pool (11%→38% of mix)	diminishing returns: median fire timing improves only $-13.1 \rightarrow -9.6 \rightarrow -7.6$ s (still seconds early) as the dose rises
3	+ trailing-silence fire pool	streaming latency crosses zero (P50 +0.32 s) but offline fire recall regresses 100→82%; the "trade-off" is first declared
4	+ complete utterances with *and* without silence tails, both labeled fire	silence becomes optional again; a strictly intermediate point on the same frontier; frontier declared fundamental
5	+ hold pool on the *same* audio as existing fire examples	collapse: opposite labels on identical inputs; the model stops firing on meeting audio entirely (fire recall 0%)
6	row 5's hold pool rebuilt on *disjoint but identically distributed* audio	bimodal oscillation between fire-mode and no-fire-mode attractors for the whole run (Figure 7, left); no checkpoint dominates; two-checkpoint ship recommended
7	**causal relabel (§3.2):** fire iff complete *and* ≥0.3 s observed silence	monotone training, both classes at ceiling simultaneously; single checkpoint dominates every predecessor (§5)

6.2 The intervention, and the diagnosis

The experiment. The frontier's fire-recall axis died by a one-variable intervention on the *evaluation*: we appended 1.0 s of silence to each clip of the legacy offline benchmark and re-scored the historical checkpoints (Table 8). The "broken" opposed-pools model's fire recall (row 6 of Table 7) rose from 0.10 to 1.00 and the mixed-pools model's (row 3) from 0.82 to 0.98, while the offline-labeled composition — which never depended on hearing silence — was the unchanged control. Under deployment-realistic input, all three fire essentially perfectly on complete utterances. The capability axis that four compositions of data engineering had been fighting was an artifact of where the evaluation clips ended.

The diagnosis. Why did the clips end where they did? Offline ASR datasets are segmented by forced alignment, so clips are cut at the end of speech, and evaluation clips inherit the cut. An offline endpoint benchmark built on such clips therefore demands a fire on audio that ends *during or immediately after* speech — a condition deployment never presents, because on a live channel silence keeps arriving after a real speaker stops. The streaming pools taught the opposite, correctly: never fire before silence is observed. The condition "complete utterance,

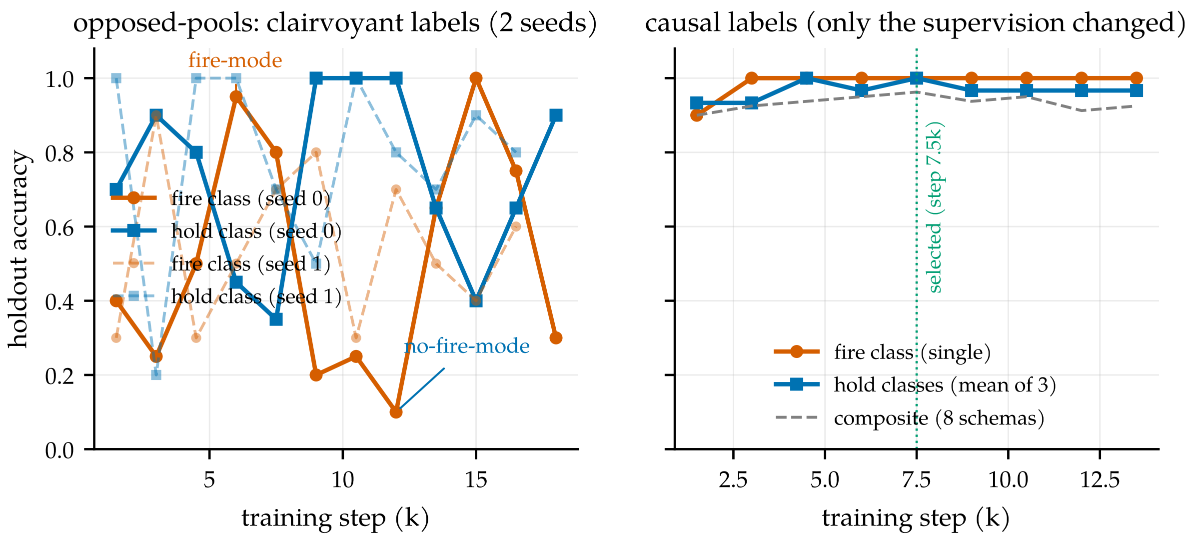

Figure 7. The fingerprint of contradictory supervision. Left: under the opposed-pools composition, holdout accuracy oscillates for the entire run between a fire-mode and a no-fire-mode attractor, reproducibly across two training seeds (solid, dashed) — the oscillation is a property of the labels, not a seed. Right: the causal recipe of §3.2 — same architecture, adapter, and data volume, only the labels changed — climbs monotonically, with both classes at ceiling simultaneously from step 3k on.

Table 8. The silence-append intervention. Fire recall on complete single utterances from the legacy offline benchmark, before and after appending 1.0 s of silence to each clip.

Model (supervision composition)	original clips	+1.0 s silence	Δ
offline-labeled (rows 1–2)	1.00	1.00	0.00
mixed-pools (row 3)	0.82	0.98	+0.16
opposed-pools (row 6)	0.10	1.00	+0.90

no trailing silence yet" thus carried the label *fire* in one pool and *hold* in the other (Figure 8a) — in the eighth composition, 50/50 label noise on the exact decision the model exists to make. A second mechanism had the same structure: the historical pools separated "disfluent pause, hold" from "turn boundary, fire" using the transcript's segmentation — same speaker continues, disfluency; another speaker follows, boundary. Measured at the decision point, mid-pause, the two classes are statistically indistinguishable: their silence-gap distributions overlap almost completely (medians 1.39 s vs. 0.83 s, and the disfluent pauses are the longer ones; Figure 8b). The label is a function of who spoke *next* — of the future. Both mechanisms are exactly the clairvoyance of §3.1, and nothing about either is specific to speech.

Why oscillation, specifically. Rows 5 and 6 of Table 7 form a natural two-condition experiment on how a contradiction expresses itself: row 6 swapped row 5's pool out rather than stacking on it, so the two conditions differ only in where the contradictory hold labels sit. When opposite labels sit on literally identical audio (row 5), the conflict is visible to the loss on every example, and the model settles deterministically on one side: it simply stops firing. When opposite labels sit on disjoint but identically distributed audio (row 6), no single example exposes the conflict; each pool is separately learnable, incompatible only in aggregate, and training circulates between mode attractors that each satisfy one pool. The oscillation is

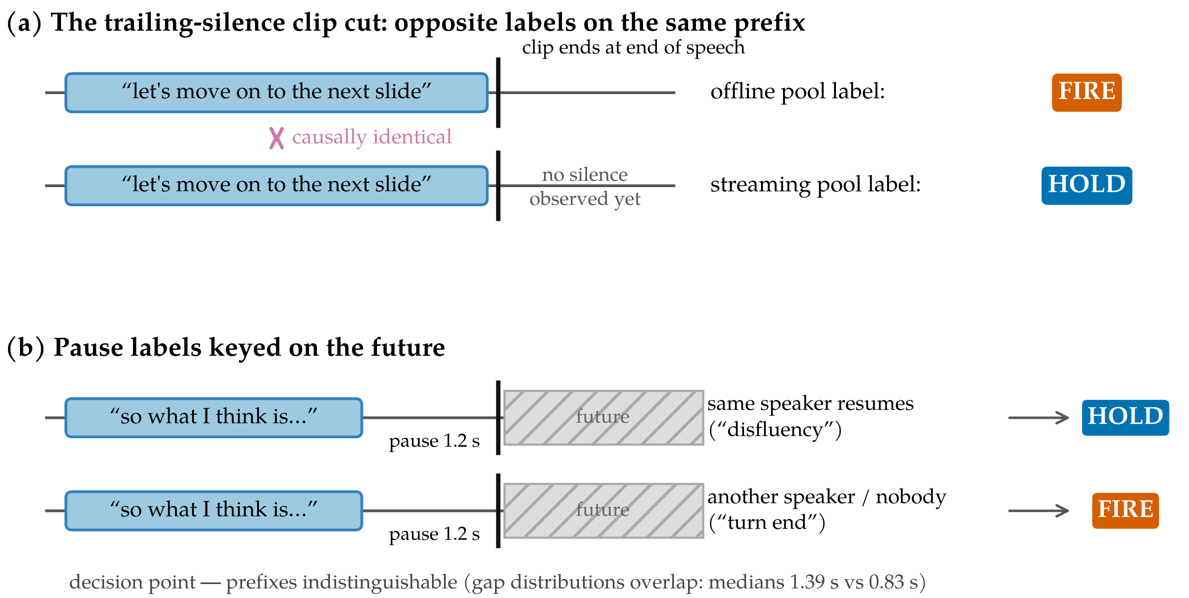

Figure 8. Two ways offline-derived supervision leaks the future. (a) Offline clips inherit forced-alignment cuts at end of speech, so the offline pool demands a fire on exactly the prefix the streaming pool labels *hold*. (b) Disfluency-vs-boundary pause labels are keyed on who speaks next; at the decision point the two classes' silence-gap distributions overlap almost completely.

not instability to be scheduled away. It is the loss surface reporting a contradiction that no individual example states. A logit-level check locates the circulation in the weights rather than at a decision threshold (Figure 11; method in Appendix D): the model's marker-commit probability swings between snapshots in lockstep with the argmax (r=0.98 on the contested fire schema). No held-out example sits persistently in the ambiguous $[0.2, 0.8]$ band across the run, and 7 of 40 fully reverse between confidently held and confidently fired — the signature of a threshold flapping at a stable probability is absent.

6.3 The mechanism, isolated

The failure reproduces outside speech, with no acoustics and no pretraining (full setup in Appendix E). A two-layer, ~30k-parameter transformer is trained on a synthetic stream in which phrase completeness is always readable from the prefix while each gap's outcome is knowable only after the decision point, and a tunable fraction f of the labels is made clairvoyant by construction. At f=0 training is textbook monotone — both classes reach 1.00, zero mode flips — and every checkpoint collapses to a single point at the top-right of the recall–precision plane. As f grows, the fingerprint develops in order (Figure 9): fire- and hold-class accuracy become anti-correlated (-0.3 at f=0.1, -1.0 at f=1.0) and checkpoints spread into an anti-diagonal cloud along recall + precision ≈ 1 — a phantom Pareto frontier that is in fact one model swinging, sampled at different steps. True bistable mode-flipping onsets only at high fraction ($f \gtrsim 0.85$), and the threshold is itself informative: because clairvoyant corruption of a given prefix is ~50/50, the causal label remains the per-prefix *majority* for every f<1, so a calm single-mode optimum survives until the contradictory pool is large enough to destabilize it. The speech model of §6.1 lived in the high-f regime by construction — its offline and streaming pools were near-balanced and directly opposed — and near-balanced opposed

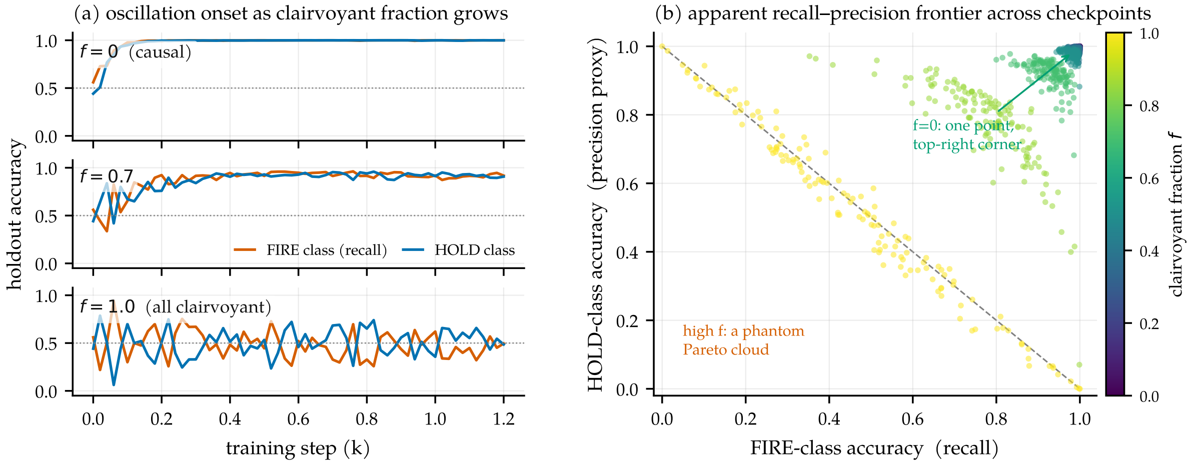

Figure 9. The mechanism, isolated. A ∼30k-parameter transformer on a synthetic stream with a tunable clairvoyant-label fraction f. (a) Holdout trajectories: monotone convergence at $f{=}0$, visible noise at $f{=}0.7$, bistable mode-oscillation at $f{=}1.0$. (b) Every checkpoint's (fire recall, hold precision), colored by f: the causal run collapses to one top-right point; high f spreads checkpoints into an anti-diagonal cloud — a phantom Pareto frontier traced by one oscillating model.

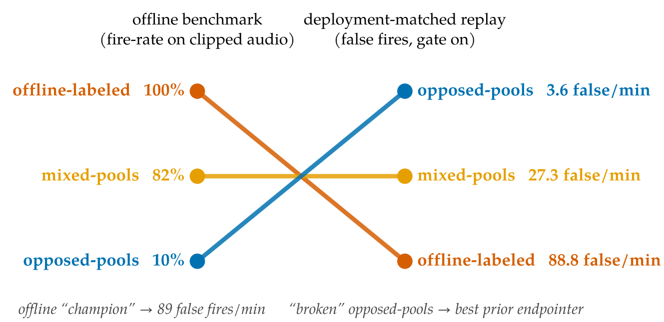

Figure 10. The ranking inversion. Model ranking on the legacy offline benchmark (left) versus the deployment-matched replay benchmark (right). The offline champion is the worst live model.

pools are exactly what mixing offline supervision with streaming reality produces.

6.4 Evaluation is where the leak enters

If clairvoyant labels are the disease, evaluation is its port of entry: the legacy offline benchmark did not just mis-score the historical models, it steered four compositions of data engineering toward a frontier that does not exist. The deployment-matched protocol of §4.1 does not merely re-score the historical models; it re-*ranks* them (Figure 10). The offline champion fires 89 times per speech-minute once audio keeps flowing — its offline strength *is* a premature-fire pathology — while the opposed-pools model, scored at 10% offline recall, turns out to be the best prior endpointer. Both historical selection decisions were wrong, and any team selecting a turn-aware checkpoint on clip-cut offline evaluations should expect the same inversion.

7 Discussion

A checklist. The failure mode generalizes to any model that must act mid-stream on offline-labeled data. Four questions for a streaming supervision pipeline:

Is your supervision clairvoyant?

1. **Prefix test.** For each label at decision time t: is it computable from input up to t? *(Our pause labels were keyed on the next speaker.)*
2. **Twin test.** Can two causally identical prefixes receive different labels anywhere in the data? *(Same complete-utterance prefix: fire in one pool, hold in the other.)* Minimal pairs differing in an *observable* are the fix, not the bug.
3. **Phantom-condition test.** Does the eval demand behavior on conditions deployment never produces? *(Clips cut at end-of-speech; deployment always delivers the next silence.)*
4. **Oscillation test.** Does training bounce between mode attractors that each satisfy part of the data? That is the loss surface reporting a contradiction.

Who else is likely affected. Candidates: turn-taking policies for full-duplex agents trained from offline dialogue transcripts, whose labels encode who *did* speak next rather than what was knowable (the projection objectives of Ekstedt and Skantze, 2022 are explicitly predictive and thus exempt); barge-in and endpointing in commercial stacks; simultaneous translation trained with reference-aligned read/write decisions [Ma et al., 2019]; streaming TTS interruption; and tool-call timing for agents cloned from completed trajectories — a streaming sibling of causal confusion in imitation learning [de Haan et al., 2019]. In each, an apparent aggressiveness-versus-caution frontier may partly be an artifact of asking the model to know the future. The diagnosis joins a lineage of measurement-induced phenomena: as sharp metric choices can manufacture "emergence" [Schaeffer et al., 2023], non-causal label choices can manufacture trade-offs.

Limitations. Single-speaker, single-channel English; the mixed-channel meeting shape is future work. The dictation capability transfers zero-shot to shorter enumerations (street addresses: 1.00 on-time recall over 60 held-out items) but not to longer-than-trained ones: on 16-digit card numbers the model fires at its learned ∼10-digit completeness point, ∼3 s early, at zero recall, with transcription accuracy unaffected throughout — a learned completeness judge remains the general lever beyond pattern-specific schemas. The unified adapter's breadth-versus-precision and recall-versus-hallucination costs are quantified in §5.3; a turn-taking-only deployment can run the pure endpointing checkpoint. Concurrency numbers are contention-caveated lower bounds from a shared GPU (Appendix G).

8 Related work

Endpointing and end-of-turn detection. Classical endpointing thresholds a VAD [Silero Team, 2021] with a silence timeout, optimized since Raux and Eskenazi [2008]; learned refinements predict end-of-query from acoustics [Shannon et al., 2017], jointly train an endpointer with the recognizer [Chang et al., 2019, Li et al., 2020, Bijwadia et al., 2023], or distill tiny acoustic end-of-turn models [Helwani et al., 2026, Pipecat team, 2025]. Recent systems fuse acoustic and linguistic cues [Wang et al., 2026, Yang et al., 2026] or fold VAD, turn detection, and ASR into one audio-LLM front-end [Li et al., 2026], the latter architecturally closest to us but with no released weights or recipe. Deployed open detectors run beside the recognizer: Smart Turn (acoustic, fully open recipe, no transcript; Pipecat team, 2025), LiveKit's turn detector and TEN (end-of-utterance classifiers over a separate STT's output; LiveKit, 2025, TEN Team, 2025, LiveKit, 2026). Open-weights streaming recognizers with in-model turn signals exist without training accounts: Kyutai STT's semantic-VAD head [Kyutai Labs, 2025] and NVIDIA's Parakeet-EOU token [NVIDIA, 2026] publish weights but neither the labeling procedure nor the data construction, and neither measures context biasing. Commercially, Deepgram Flux [Deepgram, 2026], OpenAI's semantic VAD [OpenAI, 2025], and AssemblyAI's intelligent endpointing [AssemblyAI, 2025] publish no training details. Our contribution to this line is the missing artifact — the recipe — plus evidence that recall-versus-precision trade-offs this literature reports as intrinsic can be manufactured by non-causal supervision.

Turn-taking in dialogue. Turn-taking prediction has a long tradition [Skantze, 2021], from text-based end-of-turn prediction [Ekstedt and Skantze, 2020] to voice-activity projection trained on future windows [Ekstedt and Skantze, 2022]; the human timing evidence comes from Stivers et al. [2009] and Levinson and Torreira [2015]. Projection objectives predict distributions over the future and are causally honest; our diagnosis concerns the sharper failure where future-dependent quantities are presented as deterministic labels.

Streaming and unified ASR. Streaming recognition spans RNN-T [Graves, 2012], large weakly supervised models [Radford et al., 2023], decoder-only streaming with latency losses [Wan et al., 2026], and unified streaming/offline speech-LLMs [Shi et al., 2026]. Kyutai's delayed-streams models add a semantic-VAD head to streaming STT [Kyutai Labs, 2025, Défossez et al., 2024]; full-duplex speech-to-speech models [Défossez et al., 2024] dissolve the turn abstraction entirely and are a different product class. We inherit the unified-ASR substrate and add the in-transcript turn event plus the training recipe.

Context biasing. Contextual ASR ranges from attention-based deep context [Pundak et al., 2018] to retrieval-scale biasing [Gong et al., 2025], RL-tuned hotword integration [Kong et al., 2025], and cue-based prompting for speech LLMs [Novitasari et al., 2026]; Contextual Earnings-22 [Durmus et al., 2026] concurrently standardizes entity-level evaluation on the corpus we use. We use plain prompt-slot biasing and contribute measurements on the open unified-model class.

Serving. vLLM's paged KV [Kwon et al., 2023] carries our serving path; attention sinks [Xiao et al., 2024] and bounded-state streaming serving [Li, 2026] provide the in-engine efficiency variant of our pinned context prefix.

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A Label-specification details

Completeness classifier for pair examples (Table 2, #4–5): utterance A is *incomplete* iff its transcript ends in a filler/connective ("and, but, so, or, uh, um, the, a, to, of, in, with, i mean, you know, ..."), ends mid-word (AMI partial-word annotation), or has fewer than 3 words outside a closed acknowledgment list. Borderline cases land in the `resume_after_fire` cost knob — they never create fire/hold contradictions on identical observables. Abandonment (incomplete speech, speaker never resumes) is an inference-policy timeout, not a label. Pair examples use both same- and different-speaker AMI pairs: the fire decision keys on completeness + silence, never on speaker identity, which single-channel deployment cannot observe.

B Benchmark specification and the corrected classifier

Full protocol: 0.5 s chunks; committed-prefix incremental decoding (emitted text committed, last 5 tokens rolled back) matching the base model's official streaming semantics; fires timestamped at the chunk boundary where `<END_SPEECH>` first appears; recall window $[-0.25, +1.5]$ s clipped at the next utterance's onset. The first version of the boundary classifier promoted a boundary to turn-final when another meeting speaker interleaved before the target speaker resumed. On a single-channel stream that speech is silence — demanding a fire there is exactly the clairvoyance the benchmark exists to remove. The corrected classification moved the mixed-pools model's recall 0.917→0.906 and the opposed-pools model's 0.935→0.938; no conclusion changed, but the episode is a working example of the §7 checklist applied to one's own evaluation.

C Additional results

Gated arms and the confirm-horizon sweep. Table 9. The confirm horizon behaves as a clean latency-vs-false-fire dial: h=1 strictly improves the mixed-pools model (recall *rises* because deferring the fire lets the segment gather the boundary inside tolerance) and zeroes the opposed-pools and causal models' false fires; h=2 over-suppresses (mixed-pools recall 0.812, latency past budget). The causal model is the only checkpoint whose h=0 row is already deployable. Ungated arms are omitted as uninterpretable (Figure 5): with ∼1.9 spurious fires per second on silence, the pre-diagnosis models score 0.95–0.99 "recall" by chance.

Table 9. All gated arms on the 25-stretch development benchmark. Resume = `resume_after_fire` rate over continuation boundaries (a cost knob, not an error rate). WER is a fraction, unbounded above through insertions.

Arm	Recall	P50	P95	False/min	Resume	WER_{mean}
offline-labeled + gate	0.896	0.05 s	0.25 s	88.8	0.89	0.85
mixed-pools + gate	0.906	0.16 s	0.89 s	27.3	0.96	0.36
mixed-pools + gate + confirm h=1	0.927	0.68 s	1.27 s	3.2	0.85	0.36
mixed-pools + gate + confirm h=2	0.812	1.15 s	1.42 s	1.1	0.56	0.36
opposed-pools + gate	0.938	0.26 s	0.56 s	3.6	0.93	4.96
opposed-pools + gate + confirm h=1	0.938	0.77 s	1.06 s	0.0	0.85	4.96
causal + gate	0.969	0.39 s	0.70 s	0.3	1.00	1.43
causal + gate + confirm h=1	0.969	0.89 s	1.20 s	0.0	0.93	1.43

The full timeout family. Table 10, both detectors on the development (25 stretches) and fresh confirmation (50 stretches) sets. The family's shape is stable across sets: recall collapses between X=1.0 and 1.5 s (the fire lands outside the $+1.5$ s tolerance), and no setting reaches the causal model's (recall, latency, false-fire) point. Silero [Silero Team, 2021] ran per-chunk with state resets — a slight handicap; RMS shares the LM arms' exact detector and is the apples-to-apples policy comparison (on this close-talk channel the Silero cascade is dominated by plain RMS).

Table 10. Silence-timeout cascade, all settings, on the development (dev) and fresh confirmation (conf) stretch sets.

Detector	Set	X (s)	Recall	P50	P95	False/min	Resume
RMS	dev	0.5	0.958	0.65 s	0.94 s	6.3	1.00
RMS	dev	1.0	0.969	1.15 s	1.44 s	1.4	0.81
RMS	dev	1.5	0.208	1.38 s	1.49 s	0.8	0.07
RMS	dev	2.0	0.031	0.55 s	0.55 s	0.2	0.00
RMS	conf	0.5	0.978	0.70 s	0.95 s	4.6	0.99
RMS	conf	1.0	0.989	1.20 s	1.45 s	0.6	0.73
RMS	conf	1.5	0.185	1.44 s	1.50 s	0.4	0.07
RMS	conf	2.0	0.016	1.14 s	1.14 s	0.2	0.00
Silero	dev	0.5	0.844	0.54 s	0.88 s	13.5	0.96
Silero	dev	1.0	0.854	1.04 s	1.38 s	5.5	0.85
Silero	dev	1.5	0.417	1.24 s	1.49 s	2.5	0.41
Silero	dev	2.0	0.125	1.32 s	1.42 s	1.0	0.00
Silero	conf	0.5	0.837	0.51 s	0.83 s	12.3	0.88
Silero	conf	1.0	0.864	1.00 s	1.33 s	4.4	0.73
Silero	conf	1.5	0.451	1.29 s	1.48 s	2.0	0.27
Silero	conf	2.0	0.130	1.34 s	1.44 s	1.3	0.07

Fresh-set confirmation. The causal model with the gate on the confirmation set: recall 0.924, P50 0.42 s, P95 0.70 s, 0.21 false fires per speech-minute, resume 0.91, WER median 0.30 / mean 4.63 (three long-monologue stretches enter the no-flush repetition loop; the force-flush of Appendix G is the documented fix). The 4.5 pp recall drop from the development set (93/96 vs. 170/184 turn-final boundaries) is consistent with sampling variation: the two-proportion 95% interval on the difference, $[-0.7, +9.7]$ pp, includes zero.

Probe confidence intervals and the dictation scoring window. On the enlarged probes, the released model's 95% intervals are: premature fires per sequence 0.16 $[0.12, 0.22]$, final recall 0.88 $[0.84, 0.92]$, email-with-context 0.93 $[0.87, 0.97]$, intrusion 0.008 $[0.002, 0.030]$; and on the 100-stretch replay set, boundary recall 0.982 $[0.965, 0.995]$ at 1.30 $[0.91, 1.71]$ false fires per speech-minute. The shipped scoring window books any fire within $+0.25$ s of the last speech end as premature and denies it final-recall credit. Re-scoring every fire by its offset δ from the true end of speech (genuine mid-sequence premature: $\delta < -0.30$ s; on-time: $-0.30 \leq \delta \leq +1.75$ s; late: $\delta > +1.75$ s) gives, on the same 250 sequences: 0.07 genuine mid-sequence fires per number (18/250 sequences), 0.968 on-time final recall, zero late fires, and end-fire latency P50 0.44 s (fires quantized up to the 0.5 s chunk grid). The residual under the shipped window is therefore eagerness within one chunk of the true end, not lateness; we report the shipped window in Table 4 for comparability across rows and note that it is conservative for a low-latency endpointer.

Seed robustness of the released recipe. Table 11. Retraining the released recipe (rank-32 adapter, unified ∼20k pool with 20% plain-ASR replay and natural-speech biasing, weight-decay and cosine fixes) under three seeds and selecting each best checkpoint on the probe axes gives stable premature-fire, context, and intrusion numbers — the unified capability is a property of the recipe, not a lucky checkpoint.

Table 11. Released recipe under three training seeds (best checkpoint each, smaller common probe set; the released checkpoint's enlarged-probe figures — email 0.93, intrusion 0.8% — are in Table 4). Seed 0 is the released checkpoint.

Seed	Premature/seq ↓	Email +ctx ↑	Intrusion ↓
0 (released)	0.16	0.88	0.000
1	0.16	0.98	0.000
2	0.16	0.85	0.000
mean ± sd	0.16 ± 0.00	0.90 ± 0.07	0.000 ± 0.000

Offline WER regression. Table 12. The fine-tune costs +1.1/+2.7 pp on the full official LibriSpeech splits (single-shot decode on vLLM, jiwer/whisper normalization). "Fires" is the fraction of clips where the model emitted the endpoint marker; offline clips end at or near end-of-speech, and the base model never fires because its marker rows are untrained. Banning the markers at decode time zeroes the fires without moving WER, which rules out decode truncation as the cause; Appendix F attributes the regression to narrow-schema drift and quantifies the plain-ASR-replay mitigation. The released unified model — which folds a 20% plain-ASR-replay fraction into a rank-32 adapter (§3.3) — recovers most of the gap, scoring 3.5%/6.9%.

Table 12. Offline WER (LibriSpeech, full official splits).

	WER ↓		Fires	
Model	clean	other	clean	other
Qwen3-ASR-0.6B (base)	2.79%	5.14%	0%	0%
+ endpoint LoRA (merged, causal, endpoint-only)	3.93%	7.87%	4.7%	5.9%
+ markers banned at decode	3.94%	7.91%	0%	0%
released unified (r32, +replay)	**3.49%**	**6.90%**	3.2%	4.7%

D The logit-level oscillation probe

Figure 11 is the measurement behind the mode-circulation claim of §6. The trained marker pathway is a rigid two-token sequence (<EAGER_END_SPEECH> then <END_SPEECH>) whose internal bigram is deterministic once entered, so we define P_{fire} as the marginal probability that the continuation enters the marker path rather than terminating, evaluated at the end-of-audio decision position with the transcript teacher-forced (marginalizing over the first continuation token at $\geq$0.98 probability coverage and greedy-rolling each branch). On the balanced holdout of the opposed-pools run — the exact examples behind Figure 7 (left) — mean P_{fire} tracks the recorded argmax accuracy ($r=+0.98$ on the contested fire schema; $r=-0.95$ against hold-class accuracy) and swings with amplitude $\sim$0.4 across snapshots. The per-example distribution rules out threshold flapping: not one of the 40 held-out examples sits in the ambiguous $[0.2, 0.8]$ band across all eleven snapshots (only 13–15% sit there at any single snapshot), and 7 of 40 fully reverse between confident fire ($>$0.8) and confident hold ($<$0.2) as training proceeds. The trailing-silence schema, on which both pools agree, stays pinned at $P_{\text{fire}}\approx$1.0 throughout: the circulation is confined to the schemas the pools contest.

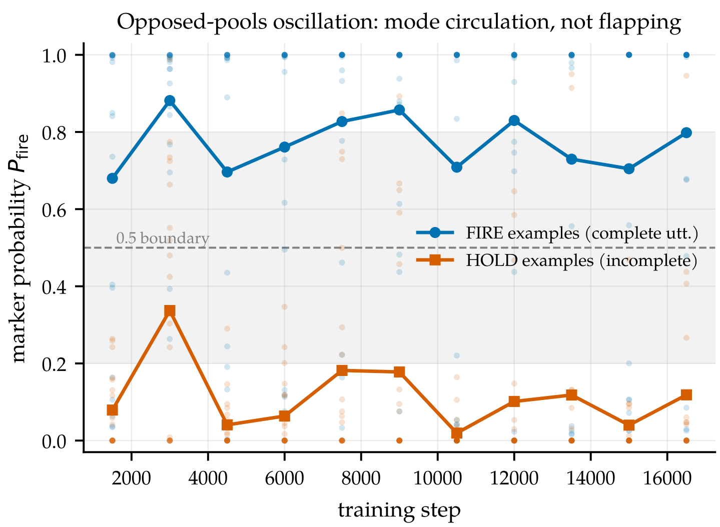

Figure 11. Mode circulation, not threshold flapping. Mean marker-commit probability P_{fire} for fire-labeled (complete utterance) and hold-labeled (incomplete) holdout examples across the opposed-pools run; small dots are per-example values. The two classes swing in lockstep as the weights orbit between the fire-mode and no-fire-mode attractors; no example hovers persistently at the 0.5 decision boundary.

E The synthetic clairvoyant-fraction experiment

Setup for the §6.3 experiment: a discrete stream of token phrases separated by silence runs, with no acoustics and no pretraining. A phrase ends complete or incomplete — always readable from the prefix — while each gap's full length (speaker resumes vs. turn ends, 50/50) is knowable only after the decision point. At a fixed depth into every gap, a two-layer causal transformer ($\sim$30k parameters) must emit fire or hold. The *causal* rule fires iff the phrase was complete; the *clairvoyant* rule fires iff the gap turns out long — a genuine future dependence, the direct analog of clips cut at boundaries and transcripts recording who spoke next. Completeness and gap length are independent, so the two rules are maximally opposed while each remains internally consistent, reproducing the "internally consistent pools with incompatible optima" structure of §3.1. A fraction f of training sequences is clairvoyantly labeled; the holdout is always causal; three seeds per f; everything runs on CPU in minutes.

The dose–response in detail (Figure 9): at f=0, both classes reach 1.00, zero mode flips, worst post-warmup holdout accuracy 0.99. The fire/hold accuracy anti-correlation grows monotonically (-0.3 at $f{=}0.1$, -0.5 at $f{=}0.5$, -1.0 at $f{=}1.0$), while true bistable mode-flipping onsets only at high fraction: 8 flips per run at $f{=}0.85$ and 16 at $f{=}1.0$, zero at $f{\leq}0.7$.

F Training-recipe notes

The unified model (§3.3) uses the same LoRA fine-tune as the rest of the paper, plus two AdamW hygiene fixes and one negative result. All three were isolated on identical data (the unified pool), so the comparisons below vary only the optimizer setting.

A weight-decay bug worth fixing, but not the WER culprit. We reserve two vocabulary rows as markers and train them by unfreezing `embed_tokens/lm_head` and masking gradients to those two rows. But AdamW's *decoupled* weight decay applies $p \leftarrow p - \eta\lambda p$ to *every* row regardless of gradient, so at $\eta{=}2{\times}10^{-4}$, $\lambda{=}0.01$ the entire (tied) 151k-row matrix shrinks $\sim$2.4% over 12k steps — silently degrading the base model's vocabulary and output head. This is a general hazard for any recipe that adds a few trainable vocabulary rows to a frozen embedding, and we fix it with a $\lambda{=}0$ group. The negative half: in a controlled comparison on identical data, the fix (plus cosine schedule) did *not* lower offline WER — the fixed unified checkpoint scores 3.9%/7.3% (clean/other) versus 3.6%/7.0% without it, within checkpoint-selection noise and in the wrong direction. The offline-WER regression is therefore *not* caused by the weight-decay bug; it is narrow-schema drift from fine-tuning on $\sim$12k examples with no plain-ASR replay data. We keep the fix for its base-model-preservation rationale, not for a WER win we cannot demonstrate.

Plain-ASR replay: the fix that does move WER. The released model mixes $\sim$4k plain transcription examples (audio $\rightarrow$ transcript, no marker, no context) into the $\sim$20k-example pool — a 20% fraction. They teach the model that a complete utterance with no context is simply to be transcribed, recovering offline WER from 3.9%/7.3% (recipe fixes, no replay) to 3.5%/6.9% and cutting spurious offline marker fires from $\sim$10–20% to 3.2%. The replay fraction trades against endpointing sharpness at fixed adapter capacity; raising the LoRA rank from 16 to 32 buys back that sharpness, which is why the released checkpoint is rank-32 (§5.3).

Cosine decay. A cosine schedule from $2{\times}10^{-4}$ to 10^{-5} after a 100-step warmup gives the most stable, highest holdout-composite trajectory (plateau 0.992 from step 6k) and a slightly less trigger-happy final checkpoint than the constant schedule.

Muon underperforms AdamW here. We tried Muon [Jordan et al., 2024] on the 2-D LoRA factors (Newton-Schulz-orthogonalized momentum, AdamW retained for the marker rows) at a $100\times$ larger base LR ($2{\times}10^{-2}$, since the orthogonalized update is $\sim$unit-norm). On identical data Muon's raw loss descends *faster* early but its task metric plateaus at holdout composite 0.846 (by step 3–4.5k, conversational schema collapsing) versus AdamW's 0.985. The cause is structural: rank-16 LoRA factors (1024×16) are too rectangular for the 5-step Newton-Schulz iteration to orthogonalize (empirically $\|O^\top O - I\|$ off-diagonals $\approx$0.5), and orthogonalizing the two factors separately is not the same operation as orthogonalizing their product. We keep AdamW.

G Serving details

Three findings from running the model on shared-GPU vLLM [Kwon et al., 2023] rather than a simulator (Figure 12), summarized in §5.5. The natural flat-cost serving shape — appending each 0.5 s chunk as a new audio placeholder in a resident request — is out of distribution for a model trained on single segments and destroys it (88% WER); bounded re-feed of the last window is the correct primitive (3.8%). The vLLM path reproduces the endpoint behavior at 84 ms median per chunk, 5.4× faster than the transformers simulator used during development — latency claims made on $O(T^2)$ re-decode simulators overstate the cost of this approach by that factor.

Bounding every resident state matters: a 20 s force-flush fixes both the WER tail and the compute tail, and per-frame P95 stays flat through 8 concurrent sessions on a 0.15-GPU slice (a contention-caveated lower bound; the shape, not the constant, is the result). Re-measured on the released rank-32 checkpoint — architecturally identical to the endpoint model after merging — serving is unchanged: median per-chunk compute is 21 ms on an otherwise-idle GPU (the 84 ms above is under a co-tenant), and per-frame P95 stays flat through 16 concurrent sessions (77 ms at N=16, well inside the 500 ms budget).

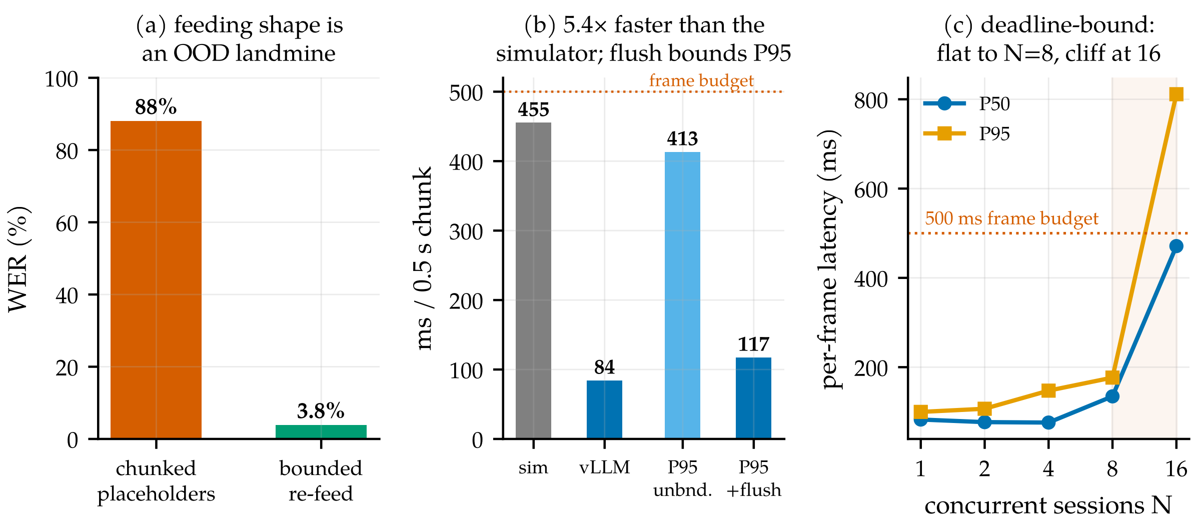

Figure 12. Serving on real infrastructure. (a) Feeding shape: per-chunk placeholders are out-of-distribution; bounded re-feed is correct. (b) Per-chunk compute on vLLM vs. the transformers simulator; the force-flush bounds the P95 tail. (c) Concurrent sessions per engine on a 0.15-GPU slice: flat to N=8, cliff at 16.

H Reproducibility

Hardware: one RTX Pro 6000 (Blackwell), shared with an unrelated co-tenant workload throughout (all GPU jobs wrapped in retry/resume; the causal training run survived four OOM kills via checkpoint resume with ≤750 steps lost each). Training: LoRA on the q/k/v/o projections plus the two marker embedding rows (endpointing experiments r=16; released unified checkpoint r=32; α=2r), batch 8, AdamW with the embedding/head weight-decay and cosine-schedule fixes of Appendix F; the unified checkpoint is selected at step 9k on the probe axes (dictation, context, intrusion) rather than the holdout composite alone. Evaluation and serving: vLLM 0.19, `skip_special_tokens=False` (the markers are reserved special tokens and are otherwise silently stripped — a gotcha worth one sentence). Code, the label-spec builder, the dictation/spelled probes, the replay benchmark, and the merged unified checkpoint are released at https://github.com/19PINE-AI/turn-aware-asr; the repository README documents the mapping from its internal checkpoint directory names to the recipes described in this paper.